Day 2: Defect Configuration Analysis

Hydrogen-Induced Defect Transport Project

Prepared for Dr. Başak Ekinci

May 21, 2026

Abstract

This document provides a detailed Day 2 plan for the project *Hydrogen-Induced Defect Transport*. Day 1 established the first one-dimensional tight-binding lattice Hamiltonian, introduced hydrogen-like defects as local on-site perturbations, computed the energy spectrum, evaluated localization using the inverse participation ratio, and generated the first dataset. Day 2 extends this foundation by systematically analyzing how defect number, defect strength, and spatial arrangement influence spectral shifts, localization, and a transport-related descriptor. The document is written in an Overleaf-ready format and includes scientific explanations, equations, Python code, terminology tables, a checklist, and references.

English Correction

The original request can be written more naturally as:

> *"Now, can you please write Day 2 in the same format as Day 1, meaning in* `.tex` *format? Please make sure everything is clear and scientifically correct."*

1 Connection with Day 1

On Day 1, we built the first simplified physics model:

clean lattice → hydrogen-like defects → energy spectrum → IPR localization → transport descriptor → first dataset.

The Day 1 model introduced hydrogen-induced defects as local changes in the on-site energy of a one-dimensional tight-binding Hamiltonian:

$$\epsilon_i \to \epsilon_i + V_H, \tag{1}$$

where ϵ_i is the on-site energy at lattice site i and V_H is the hydrogen-like perturbation strength. Day 2 now asks a more systematic question:

> **How do the number, strength, and spatial arrangement of hydrogen-induced defects affect the energy spectrum, localization, and transport-related descriptors?**

2 Day 2 Goal

The goal of Day 2 is to move from a single randomly defected lattice to controlled defect configuration experiments.

By the end of Day 2, we should have:

1. a new notebook named `02_defect_configuration_analysis.ipynb`,
2. controlled defect configurations,
3. clean versus defected energy spectrum comparison,
4. spectral shift calculation,
5. defect number sweep,
6. defect strength sweep,
7. random, clustered, and spread defect arrangement comparison,
8. plots showing changes in localization,
9. plots showing changes in the transport descriptor,
10. a Day 2 dataset saved as `day2_defect_configuration_dataset.csv`.

3 Scientific Motivation

In Day 1, defect positions were generated randomly. This was useful for producing a first dataset, but it does not yet tell us which physical factor is most important.

In Day 2, we separate the effects of:

1. defect number,
2. defect strength,
3. defect arrangement.

These variables modify the clean Hamiltonian:

$$H_{\text{clean}} \rightarrow H_{\text{defected}}. \tag{2}$$

Because the Hamiltonian determines the quantum states, any change in the Hamiltonian can modify the eigenvalue problem:

$$H_{\text{defected}}\psi_n = E_n\psi_n. \tag{3}$$

Therefore, Day 2 is a controlled physics experiment. Instead of changing everything at once, we change one factor at a time and observe the effect on the spectrum, localization, and transport descriptor.

4 Main Scientific Question

The central Day 2 question is:

$$\text{How sensitive are localization and transport descriptors to hydrogen defect configurations?} \tag{4}$$

In simpler language:

> Do electrons become more localized when we add more defects, stronger defects, or more clustered defects?

5 Physical Variables Studied on Day 2

5.1 Defect Number

The first variable is the number of defects:

$$N_{\text{defects}} = 1, 2, 3, \dots, 15. \tag{5}$$

Increasing the number of hydrogen-like defects introduces more disorder into the lattice. The expected qualitative behavior is:

more defects → more disorder → stronger localization → larger mean IPR → smaller transport score.

Mathematically, we expect the tendency:

$$N_{\text{defects}} \uparrow \quad \Rightarrow \quad \langle IPR \rangle \uparrow \quad \Rightarrow \quad D_{\text{transport}} \downarrow . \tag{6}$$

The transport descriptor remains the same as in Day 1:

$$D_{\text{transport}} = \frac{1}{\langle IPR \rangle}. \tag{7}$$

5.2 Defect Strength

The second variable is the perturbation strength:

$$V_H. \tag{8}$$

This controls how strongly the hydrogen-like defect changes the local on-site energy:

$$\epsilon_i \to \epsilon_i + V_H. \tag{9}$$

If V_H is small, the defect is weak. If V_H is large, the defect strongly modifies the local energy landscape.

The expected qualitative behavior is:

larger V_H → stronger local perturbation → larger spectral shift → stronger localization → lower transport descriptor.

5.3 Defect Arrangement

The third variable is the spatial arrangement of defects.

Two systems may have the same number of defects and the same defect strength, but different spatial distributions. For example:

- **Clustered defects:** sites 10, 11, and 12.
- **Spread defects:** sites 5, 20, and 35.

Clustered defects may form a strong local trapping region and increase the maximum IPR. Spread defects may disturb the lattice more globally and produce a different spectral response.

Day 2 compares three arrangement modes:

1. random defects,
2. clustered defects,
3. spread defects.

6 Clean versus Defected Spectrum

For the clean lattice, we compute:

$$E_n^{\text{clean}}, \quad \psi_n^{\text{clean}}, \quad IPR_n^{\text{clean}}. \tag{10}$$

For the defected lattice, we compute:

$$E_n^{\text{def}}, \quad \psi_n^{\text{def}}, \quad IPR_n^{\text{def}}. \tag{11}$$

Then we calculate the spectral shift:

$$\Delta E_n = E_n^{\text{def}} - E_n^{\text{clean}}. \tag{12}$$

Large values of ΔE_n indicate that the corresponding energy state is sensitive to the hydrogen-like defect configuration.

7 Day 2 Notebook Name

Create the following notebook:

`notebooks/02_defect_configuration_analysis.ipynb`

8 Day 2 Notebook Structure

The notebook should contain:

1. Scientific motivation,
2. imports,
3. Day 1 helper functions,
4. controlled defect site function,
5. Hamiltonian analysis function,
6. clean versus defected spectrum comparison,
7. spectral shift visualization,
8. defect number sweep,
9. defect strength sweep,
10. random versus clustered versus spread comparison,
11. Day 2 dataset generation,
12. dataset saving,
13. Day 2 conclusion.

9 Step 1: Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from scipy.linalg import eigh
```

9.1 Scientific Meaning

These packages are used for matrix operations, data handling, plotting, and solving the eigenvalue problem:

$$H\psi_n = E_n\psi_n. \tag{13}$$

9.2 New Terms

Term	Explanation
Import	Loading a Python package into the notebook.
Eigenvalue solver	Numerical method used to solve $H\psi = E\psi$.
DataFrame	A table-like data structure from `pandas`.
Visualization	Plotting results to interpret scientific patterns.

10 Step 2: Reuse Day 1 Functions

10.1 Clean Hamiltonian Function

```
def build_clean_hamiltonian(N, t=1.0, epsilon_0=0.0):
    """
    Build a clean 1D nearest-neighbor tight-binding Hamiltonian.
    """
    H = np.zeros((N, N))

    np.fill_diagonal(H, epsilon_0)

    for i in range(N - 1):
        H[i, i + 1] = -t
        H[i + 1, i] = -t

    return H
```

10.2 IPR Function

```
def compute_ipr(eigenvectors):
    """
    Compute inverse participation ratio for each eigenstate.

    IPR_n = sum_i |psi_n(i)|^4
    """
    return np.sum(np.abs(eigenvectors) ** 4, axis=0)
```

10.3 Scientific Meaning

The clean Hamiltonian function builds the baseline lattice without defects. The IPR function measures how localized each eigenstate is:

$$IPR_n = \sum_i |\psi_n(i)|^4. \tag{14}$$

Small IPR values indicate extended states. Large IPR values indicate localized states.

11 Step 3: Add Defects at Chosen Sites

On Day 1, defects were placed randomly. On Day 2, we introduce a controlled function that allows defects to be placed at specific sites.

```
def add_defects_at_sites(H, defect_sites, defect_strength=2.0):
    """
    Add defects to specific lattice sites.

    Parameters
    ----------
    H : np.ndarray
        Clean Hamiltonian.
    defect_sites : list or np.ndarray
        Sites where defects are placed.
    defect_strength : float
        Defect perturbation strength.

    Returns
    -------
    H_def : np.ndarray
        Defected Hamiltonian.
    """
    H_def = H.copy()

    for site in defect_sites:
        H_def[site, site] += defect_strength

    return H_def
```

11.1 Scientific Meaning

This function applies the perturbation:

$$\epsilon_i \to \epsilon_i + V_H, \tag{15}$$

but now the sites are selected by the researcher. This is important because it creates a controlled experiment.

11.2 New Terms

Term	Explanation
Controlled defect	A defect placed at a predefined site.
Perturbation strength	Magnitude of the local on-site energy change.
Defected Hamiltonian	Hamiltonian after local perturbations are added.

Controlled experiment	Experiment where one factor is changed while others are fixed.

12 Step 4: Analyze One Hamiltonian

```
def analyze_hamiltonian(H):
    """
    Compute spectrum, eigenvectors, IPR, and transport descriptor.
    """
    eigenvalues, eigenvectors = eigh(H)
    ipr_values = compute_ipr(eigenvectors)

    mean_ipr = np.mean(ipr_values)
    max_ipr = np.max(ipr_values)
    transport_score = 1.0 / mean_ipr

    return {
        "eigenvalues": eigenvalues,
        "eigenvectors": eigenvectors,
        "ipr_values": ipr_values,
        "mean_ipr": mean_ipr,
        "max_ipr": max_ipr,
        "transport_score": transport_score
    }
```

12.1 Scientific Meaning

This function calculates all key descriptors for a Hamiltonian:

1. energy eigenvalues,
2. eigenvectors,
3. IPR values,
4. mean IPR,
5. maximum IPR,
6. transport score.

The transport score is defined as:

$$D_{\text{transport}} = \frac{1}{\langle IPR \rangle}. \tag{16}$$

This is a proxy, not a true conductance calculation. It is used as a first descriptor of how extended or localized the eigenstates are.

13 Step 5: Define Basic Parameters

```
N = 40
t = 1.0
epsilon_0 = 0.0
defect_strength = 2.0
```

```
H_clean = build_clean_hamiltonian(N, t, epsilon_0)
clean_results = analyze_hamiltonian(H_clean)
```

13.1 Scientific Meaning

The lattice contains $N = 40$ sites. The hopping strength is set to $t = 1.0$, and the base on-site energy is $\epsilon_0 = 0.0$. These values are dimensionless in this toy model.

14 Part A: Clean versus Defected Spectrum

14.1 Step 6: Create One Defected System

```
defect_sites = [10, 20, 30]

H_def = add_defects_at_sites(
    H_clean,
    defect_sites=defect_sites,
    defect_strength=defect_strength
)

def_results = analyze_hamiltonian(H_def)
```

14.2 Scientific Meaning

Here, three hydrogen-like defects are placed at sites 10, 20, and 30. The goal is to compare this defected Hamiltonian with the clean Hamiltonian.

14.3 Step 7: Plot Clean versus Defected Spectrum

```
plt.figure(figsize=(8, 5))
plt.plot(
    clean_results["eigenvalues"],
    marker="o",
    linestyle="none",
    label="Clean lattice"
)
plt.plot(
    def_results["eigenvalues"],
    marker="x",
    linestyle="none",
    label="Defected lattice"
)
plt.xlabel("Eigenstate index")
plt.ylabel("Energy")
plt.title("Clean vs Defected Energy Spectrum")
plt.legend()
plt.grid(True)
plt.show()
```

14.4 Scientific Interpretation

If the clean and defected spectra differ, then the hydrogen-like defects changed the allowed energy levels. This means:

defect perturbation → changed Hamiltonian → changed energy spectrum.

14.5 Step 8: Plot Spectral Shift

```
delta_E = def_results["eigenvalues"] - clean_results["eigenvalues"]

plt.figure(figsize=(8, 5))
plt.plot(delta_E, marker="o")
plt.xlabel("Eigenstate index")
plt.ylabel(r"$\Delta E_n$")
plt.title("Spectral Shift Caused by Hydrogen-Like Defects")
plt.grid(True)
plt.show()
```

14.6 Scientific Meaning

The spectral shift is:

$$\Delta E_n = E_n^{\mathrm{def}} - E_n^{\mathrm{clean}}. \tag{17}$$

Large values of ΔE_n mean that those energy states are more sensitive to the defect perturbation.

15 Part B: Defect Number Sweep

15.1 Step 9: Sweep Number of Defects

```
def sweep_defect_number(
    N=40,
    t=1.0,
    epsilon_0=0.0,
    defect_strength=2.0,
    max_defects=15,
    seed=42
):
    np.random.seed(seed)
    H_clean = build_clean_hamiltonian(N, t, epsilon_0)

    rows = []

    for num_defects in range(1, max_defects + 1):
        defect_sites = np.random.choice(N, size=num_defects, replace=
            False)

        H_def = add_defects_at_sites(
            H_clean,
            defect_sites=defect_sites,
            defect_strength=defect_strength
        )

        results = analyze_hamiltonian(H_def)

        rows.append({
            "num_defects": num_defects,
            "defect_strength": defect_strength,
            "defect_sites": defect_sites.tolist(),
```

```
            "mean_ipr": results["mean_ipr"],
            "max_ipr": results["max_ipr"],
            "transport_score": results["transport_score"]
        })

    return pd.DataFrame(rows)
```

Run:

```
df_number = sweep_defect_number()
df_number.head()
```

15.2 Scientific Meaning

This is a parameter sweep. We change the number of defects from 1 to 15 while keeping the defect strength fixed at $V_H = 2.0$. This allows us to isolate the effect of defect number.

15.3 Step 10: Plot Defect Number versus Mean IPR

```
plt.figure(figsize=(8, 5))
plt.plot(df_number["num_defects"], df_number["mean_ipr"], marker="o")
plt.xlabel("Number of defects")
plt.ylabel("Mean IPR")
plt.title("Effect of Defect Number on Localization")
plt.grid(True)
plt.show()
```

15.4 Step 11: Plot Defect Number versus Transport Score

```
plt.figure(figsize=(8, 5))
plt.plot(df_number["num_defects"], df_number["transport_score"], marker="
    o")
plt.xlabel("Number of defects")
plt.ylabel("Transport score")
plt.title("Effect of Defect Number on Transport Descriptor")
plt.grid(True)
plt.show()
```

15.5 Expected Scientific Interpretation

If increasing the number of defects increases disorder, then localization may increase. This would appear as:

$$N_{\text{defects}} \uparrow \quad \Rightarrow \quad \langle IPR \rangle \uparrow \quad \Rightarrow \quad D_{\text{transport}} \downarrow . \tag{18}$$

However, in a finite one-dimensional toy model, the trend may not be perfectly monotonic because random positions can affect localization differently. This is not an error; it is part of the physics exploration.

16 Part C: Defect Strength Sweep

16.1 Step 12: Sweep Defect Strength

```python
def sweep_defect_strength(
    N=40,
    t=1.0,
    epsilon_0=0.0,
    num_defects=5,
    strength_values=None,
    seed=42
):
    if strength_values is None:
        strength_values = np.linspace(0.1, 5.0, 30)

    np.random.seed(seed)
    H_clean = build_clean_hamiltonian(N, t, epsilon_0)

    fixed_defect_sites = np.random.choice(N, size=num_defects, replace=
       False)

    rows = []

    for strength in strength_values:
        H_def = add_defects_at_sites(
            H_clean,
            defect_sites=fixed_defect_sites,
            defect_strength=strength
        )

        results = analyze_hamiltonian(H_def)

        rows.append({
            "num_defects": num_defects,
            "defect_strength": strength,
            "defect_sites": fixed_defect_sites.tolist(),
            "mean_ipr": results["mean_ipr"],
            "max_ipr": results["max_ipr"],
            "transport_score": results["transport_score"]
        })

    return pd.DataFrame(rows)
```

Run:

```python
df_strength = sweep_defect_strength()
df_strength.head()
```

16.2 Scientific Meaning

This sweep changes only the defect strength while keeping the defect sites fixed. This is important because it isolates the effect of V_H.

16.3 Step 13: Plot Defect Strength versus Mean IPR

```python
plt.figure(figsize=(8, 5))
plt.plot(df_strength["defect_strength"], df_strength["mean_ipr"], marker=
   "o")
plt.xlabel("Defect strength")
plt.ylabel("Mean IPR")
plt.title("Effect of Defect Strength on Localization")
```

```
plt.grid(True)
plt.show()
```

16.4 Step 14: Plot Defect Strength versus Transport Score

```
plt.figure(figsize=(8, 5))
plt.plot(df_strength["defect_strength"], df_strength["transport_score"],
    marker="o")
plt.xlabel("Defect strength")
plt.ylabel("Transport score")
plt.title("Effect of Defect Strength on Transport Descriptor")
plt.grid(True)
plt.show()
```

16.5 Expected Scientific Interpretation

If stronger defects create stronger local perturbations, localization may increase and transport score may decrease:

$$V_H \uparrow \quad \Rightarrow \quad \langle IPR \rangle \uparrow \quad \Rightarrow \quad D_{\text{transport}} \downarrow . \tag{19}$$

17 Part D: Defect Arrangement Comparison

17.1 Step 15: Define Random, Clustered, and Spread Defects

```
def generate_defect_sites(N, num_defects, mode="random", seed=42):
    np.random.seed(seed)

    if mode == "random":
        return np.random.choice(N, size=num_defects, replace=False)

    elif mode == "clustered":
        start = np.random.randint(0, N - num_defects)
        return np.arange(start, start + num_defects)

    elif mode == "spread":
        return np.linspace(0, N - 1, num_defects, dtype=int)

    else:
        raise ValueError("mode must be one of: random, clustered, spread"
            )
```

17.2 Scientific Meaning

This function creates three spatial configurations:

Mode	Meaning
`random`	Defects are randomly distributed across the lattice.
`clustered`	Defects are placed next to each other.
`spread`	Defects are distributed across the lattice.

17.3 Step 16: Compare Arrangements

```
def compare_defect_arrangements(
    N=40,
    t=1.0,
    epsilon_0=0.0,
    num_defects=5,
    defect_strength=2.0,
    seed=42
):
    H_clean = build_clean_hamiltonian(N, t, epsilon_0)

    rows = []

    for mode in ["random", "clustered", "spread"]:
        defect_sites = generate_defect_sites(
            N=N,
            num_defects=num_defects,
            mode=mode,
            seed=seed
        )

        H_def = add_defects_at_sites(
            H_clean,
            defect_sites=defect_sites,
            defect_strength=defect_strength
        )

        results = analyze_hamiltonian(H_def)

        rows.append({
            "arrangement": mode,
            "num_defects": num_defects,
            "defect_strength": defect_strength,
            "defect_sites": defect_sites.tolist(),
            "mean_ipr": results["mean_ipr"],
            "max_ipr": results["max_ipr"],
            "transport_score": results["transport_score"]
        })

    return pd.DataFrame(rows)
```

Run:

```
df_arrangement = compare_defect_arrangements()
df_arrangement
```

17.4 Step 17: Plot Arrangement versus Mean IPR

```
plt.figure(figsize=(8, 5))
plt.bar(df_arrangement["arrangement"], df_arrangement["mean_ipr"])
plt.xlabel("Defect arrangement")
plt.ylabel("Mean IPR")
plt.title("Effect of Defect Arrangement on Localization")
plt.grid(axis="y")
plt.show()
```

17.5 Step 18: Plot Arrangement versus Transport Score

```
plt.figure(figsize=(8, 5))
plt.bar(df_arrangement["arrangement"], df_arrangement["transport_score"])
plt.xlabel("Defect arrangement")
plt.ylabel("Transport score")
plt.title("Effect of Defect Arrangement on Transport Descriptor")
plt.grid(axis="y")
plt.show()
```

17.6 Expected Scientific Interpretation

Clustered defects may increase the maximum IPR by creating a stronger local trapping region. Spread defects may create a more distributed perturbation. Random defects may give intermediate or seed-dependent results. In a finite toy model, the arrangement effect may depend strongly on the chosen defect positions.

18 Part E: Generate the Day 2 Dataset

18.1 Step 19: Combine All Day 2 Experiments

```
df_number["experiment"] = "defect_number_sweep"
df_strength["experiment"] = "defect_strength_sweep"
df_arrangement["experiment"] = "defect_arrangement_comparison"

day2_dataset = pd.concat(
    [df_number, df_strength, df_arrangement],
    ignore_index=True
)

day2_dataset.head()
```

18.2 Step 20: Save the Day 2 Dataset

```
day2_dataset.to_csv(
    "../data/day2_defect_configuration_dataset.csv",
    index=False
)

print("Saved: ../data/day2_defect_configuration_dataset.csv")
```

18.3 Dataset Meaning

The Day 2 dataset contains controlled physics experiments. It is more informative than the Day 1 dataset because it records which experiment generated each result.

Column	Meaning
`experiment`	Type of Day 2 experiment.
`num_defects`	Number of hydrogen-like defects.
`defect_strength`	Local perturbation strength.
`defect_sites`	Sites where defects are placed.
`arrangement`	Random, clustered, or spread arrangement.

`mean_ipr`	Average localization index.
`max_ipr`	Strongest localization index.
`transport_score`	Inverse of mean IPR.

19 Day 2 Conclusion Text

At the end of the notebook, write:

> In this notebook, we extended the Day 1 lattice Hamiltonian model by systematically analyzing hydrogen-induced defect configurations. We compared clean and defected spectra, measured spectral shifts, and studied how defect number, defect strength, and spatial arrangement influence localization and the transport descriptor. The results provide a more controlled physics-informed dataset for later machine learning and quantum machine learning analysis.

20 New Terms for Day 2

Term	**Meaning**
Defect configuration	Specific choice of defect number, strength, and position.
Spectral shift	Energy-level change caused by defects.
Sweep	Systematic variation of one parameter.
Clustered defects	Defects placed close together.
Spread defects	Defects distributed across the lattice.
Sensitivity	How strongly a result changes when a parameter changes.
Controlled experiment	Experiment where one variable changes while others are fixed.
Transport proxy	Approximate quantity related to electron movement.
Arrangement	Spatial organization of defects.
Monotonic trend	A trend that consistently increases or decreases.

21 Day 2 Success Checklist

- ☐ I created `02_defect_configuration_analysis.ipynb`.
- ☐ I wrote the Day 2 scientific motivation.
- ☐ I reused the Day 1 Hamiltonian and IPR functions.
- ☐ I created a function for controlled defect sites.
- ☐ I compared clean and defected energy spectra.
- ☐ I plotted spectral shift.
- ☐ I performed a defect number sweep.
- ☐ I plotted defect number versus mean IPR.
- ☐ I plotted defect number versus transport score.
- ☐ I performed a defect strength sweep.
- ☐ I plotted defect strength versus mean IPR.

- ☐ I plotted defect strength versus transport score.
- ☐ I compared random, clustered, and spread defects.
- ☐ I generated the Day 2 dataset.
- ☐ I saved `day2_defect_configuration_dataset.csv`.
- ☐ I wrote the Day 2 conclusion.

22 Main Sentence for Today

Today, I studied how defect number, defect strength, and defect arrangement affect localization and the transport descriptor.

23 Mini English Practice

Useful Sentence

Today, I analyzed how different hydrogen defect configurations affect the energy spectrum and localization behavior.

Vocabulary

Word	Meaning	Example Sentence
Configuration	Arrangement or setup.	We tested different defect configurations.
Sensitivity	Degree of response to a change.	The spectrum shows sensitivity to defect strength.
Sweep	Systematic parameter variation.	We performed a defect strength sweep.
Clustered	Placed close together.	Clustered defects may create stronger localization.
Spread	Distributed across space.	Spread defects disturb the lattice globally.
Descriptor	Numerical feature.	The transport descriptor decreases when localization increases.

24 Recommended References

24.1 Tight-Binding and Lattice Hamiltonians

1. David Tong, *Applications of Quantum Mechanics*. Tight-binding discussion.
   https://www.damtp.cam.ac.uk/user/tong/aqm/aqmtwo.pdf
2. Tight-binding model notes.
   https://ssp.utasphys.cloud.edu.au/3-1d/3-3-tightbinding/
3. Kwant tight-binding introduction.
   https://kwant-project.org/doc/latest/tutorial/introduction

24.2 Eigenvalue Calculations

1. SciPy `eigh` documentation.
   https://docs.scipy.org/doc/scipy/reference/generated/scipy.linalg.eigh.html

2. NumPy `eigh` documentation.
   https://numpy.org/doc/stable/reference/generated/numpy.linalg.eigh.html

24.3 Localization and IPR

1. Inverse Participation Ratio overview.
   https://www.emergentmind.com/topics/inverse-participation-ratio-ipr

2. Anderson localization background notes.
   https://web.mit.edu/8.334/www/grades/projects/projects19/GuanChenguang.pdf

24.4 Future Quantum Transport and Quantum Machine Learning

1. Kwant tutorial for quantum transport.
   https://kwant-project.org/doc/dev/tutorial/

2. Qiskit Machine Learning quantum kernel tutorial.
   https://qiskit-community.github.io/qiskit-machine-learning/tutorials/03_quantum_kernel.html

3. IBM Quantum kernel training tutorial.
   https://quantum.cloud.ibm.com/docs/tutorials/quantum-kernel-training

25 Next Step After Day 2

After completing Day 2, the next step is Day 3:

> Perform a more detailed spectral and localization analysis by comparing clean and defected systems across multiple random seeds, then visualize the stability of the descriptors using repeated experiments.